Assignment 2

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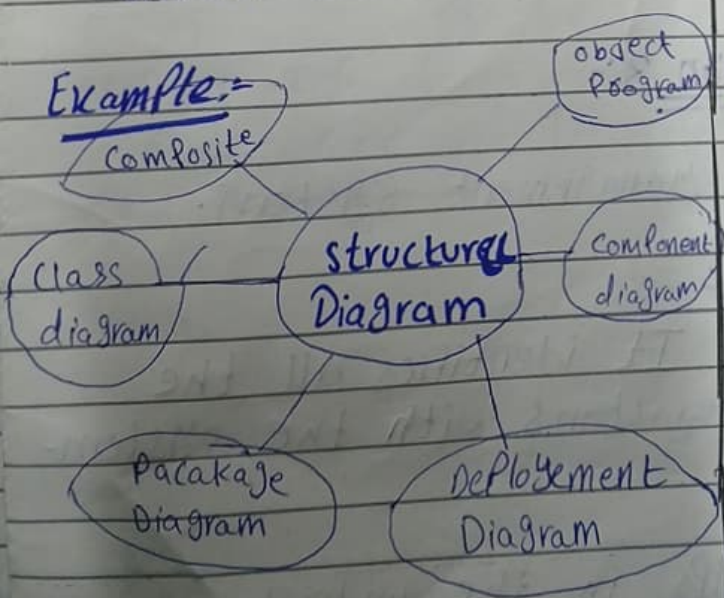
(1)

Distinguish b/w structural & Behavioural UML:-

Structural UML Diagram	Behavioural UML Diagram
(1) It show the static aspects of a system.	Show the dynamic aspects of a system
(2) It focuses on structure, components and relationship.	Focus on behaviour, inspections and state changes.
(3) Typically used by architects and devolpers.	(3) It is used by analyist and testers
(4) Represents how the system is built:	(4) Represent how the system behaves or operates.
Example:-	Example:-



- Use case diagram
- Sequence diagram
- Communication diagram
- Inspection diagram
- Activity Diagram
- State Chart Diagram

Question 2:

System Modeling in Software devolpment means creating diagrams and models to represent a systems, structure, behaviour and functionality The Process of creating abstract, representation of a system, its components and their interactions.

→ It is crucial because it:

- It improves understanding.
- Reduse complexity
- enhances ~~error~~ communication
- identifies error early.
- Supports iterative devolpment.

Question 3:-

I selected Library Management System.

1) **Use Case Diagram:-**

It identifies all the action and their interactions with the system.

2. **Contribution:-**

It helps in the understanding user requirements and outliving key functionalities such manging accounts. It is useful in requirement gathering and system planing.

2:- Class diagram:-

It defines the Static structure of the System including main classe like Book, Member, birarian and loan, along with their attributes and relation ship's.

Contributions:

Assists in designing the database Schema and object oriented structure. It ensure a Class understanding of how data is organised and their relation with system

3: Sequence Diagram:-

It shows the sequence of interactions b/w objects for Procone issuing or retrieving a book.

Contribution:-

It helps in under standing the step by step flow of operations ensuring all component Communicate correctly - This is useful during development and testing Phases to validate behaviour.